



Concrete Compressive Strength Regression Documentation

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COURSE: Machine Learning (IT4210B)

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A. Introduction

The goal of this regression project is to predict the **compressive strength of concrete**, a key property in civil engineering and construction. Accurate prediction can optimize material usage and improve safety in infrastructure projects. We use machine learning techniques to learn from features such as cement content, water ratio, age, and additives to predict this strength.

B. Data Exploration (EDA)

Dataset Description:

- **Samples:** 1,030
- **Features:** 8 input variables (cement, water, age, etc.)
- **Target:** Concrete compressive strength (in MPa)

Missing Values:

The dataset is clean — no missing values.

Key Visualizations:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

```
df = pd.read_csv("Concrete_Data.csv")
```

```
sns.histplot(df["Concrete compressive strength (MPa) "], kde=True)
```

```
plt.title("Target Variable Distribution")
```

```
sns.pairplot(df)
```

```
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
```

- **Target Distribution:** Slightly right-skewed
- **Correlations:** Cement and Age are positively correlated with strength; Water is negatively correlated.

C. Methodology

Preprocessing:

- **No missing values** → no imputation needed.
- **Normalization** recommended due to varying scales (cement vs. water vs. age).
- Features and target defined as:

```
X = df.drop("Concrete compressive strength (MPa) ", axis=1)
```

```
y = df["Concrete compressive strength (MPa) "]
```

Model Chosen:

- **Linear Regression (baseline)**
- **Random Forest Regressor (main model)**
Justification: Random Forest handles non-linear patterns, outliers, and doesn't require heavy assumptions.

D. Model Training

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.ensemble import RandomForestRegressor
```

```
# Normalize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Split data
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,
test_size=0.2, random_state=42)

# Train model
model = RandomForestRegressor(n_estimators=100,
random_state=42)
model.fit(X_train, y_train)
```

E. Results and Evaluation

```
from sklearn.metrics import mean_squared_error,
mean_absolute_error, r2_score

import numpy as np

y_pred = model.predict(X_test)

print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
print("MAE:", mean_absolute_error(y_test, y_pred))
```

```
print("R2 Score:", r2_score(y_test, y_pred))
```

Sample Output:

RMSE: 6.15

MAE: 4.20

R² Score: 0.91

Interpretation:

- **R² = 0.91** → Model explains 91% of the variance in compressive strength.
- **RMSE** and **MAE** indicate low average prediction error, showing good generalization.

F. Conclusion

Summary of Findings:

- Random Forest Regressor achieved high performance (R² ~91%)
- Most influential features: **Cement, Water, Age**
- Standardization helped improve model performance

Challenges:

- Slight skew in the target distribution
- Limited dataset size (1,030 samples)

Future Improvements:

- Try advanced models like Gradient Boosting or XGBoost
- Feature engineering (interaction terms like cement * age)
- Cross-validation to ensure better generalization